

National Infrastructure Projects

SQL Server + Power BI | Budget Variance · Cost Overrun · Equipment Risk

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Executive Summary

This portfolio project analyzes infrastructure project budgets, actual costs, cost variance, project overruns, variance reasons, and equipment maintenance activity. The SQL layer creates three reusable analytical views, while the Power BI report presents executive KPIs and interactive visual analysis.

The supplied dashboard reports **371.23 bn SAR** planned cost, **482.97 bn SAR** actual cost, a **30.10%** overrun, and **111.73 bn SAR** total variance.

1. Business Problem

Infrastructure programs need a consistent way to identify budget pressure, understand variance drivers, compare project types and regions, and prioritize equipment-related operational risk.

2. Project Objectives

- Validate the available SQL Server source tables and record volumes.
- Analyze project-level overrun performance and cost variance drivers.
- Connect detailed variance and maintenance records to project context.
- Create reusable SQL views for Power BI consumption.
- Present budget, cost, regional, project-type, monthly, and equipment signals in one dashboard.

3. Dataset Overview

Dataset	Records	Primary analytical use
01_Projects_Master	5,000	Projects, budgets, actuals, overruns, delays, status, team size
02_Cost_Overrun_Details	24,994	Variance reasons, categories, planned/actual cost, variance
03_Maintenance_Logs	29,547	Equipment, maintenance, severity, cost, downtime, status

The source dataset file itself was not included with the supplied materials; therefore this package does not fabricate or substitute an XLSX file.

4. SQL Analysis

The supplied SQL creates the ConstructionAnalytics database, checks available tables, validates row counts, previews project records, runs three business questions, creates three analytical views, tests the views, and checks the active SQL Server/database environment.

EDA & Validation

- Table-name inspection through INFORMATION_SCHEMA
- Row counts across all three source tables
- TOP 5 project-master preview
- View row-count validation
- Server, instance, and current database checks

Data Cleaning

No explicit cleaning pipeline is present in the supplied SQL. There are no NULL-removal rules, duplicate checks/removals, TRIM/REPLACE operations, type conversions, date conversions, or text standardization statements.

Analytical SQL

- COUNT, AVG, and SUM aggregations
- GROUP BY with ORDER BY
- JOINS between project, cost-overflow, and maintenance data
- CASE-based risk categorization
- DATEDIFF-based planned duration

Advanced SQL

No window functions, ROW_NUMBER, RANK/DENSE_RANK, SUM OVER, running totals, cumulative calculations, PARTITION BY, CTEs, or subqueries are present.

5. Important SQL Techniques

Project-type overrun comparison

```
SELECT Project_Type, COUNT(*) AS Total_Projects, AVG(Overrun_Pct) AS Avg_Overrun_Pct,
SUM(Cost_Overrun_SAR) AS Total_Overrun_SAR FROM [01_Projects_Master]
GROUP BY Project_Type ORDER BY Avg_Overrun_Pct DESC;
```

Variance root-cause summary

```
SELECT Reason_For_Variance, COUNT(*) AS Occurrences, SUM(Variance_SAR) AS Total_Loss_SAR
FROM [02_Cost_Overrun_Details] GROUP BY Reason_For_Variance ORDER BY Total_Loss_SAR DESC;
```

Equipment downtime analysis

```
SELECT Equipment, SUM(Downtime_Hours) AS Total_Downtime, AVG(Maintenance_Cost_SAR) AS
Avg_Cost
FROM [03_Maintenance_Logs] GROUP BY Equipment ORDER BY Total_Downtime DESC;
```

Risk-aware KPI view

```
CASE WHEN Overrun_Pct > 20 THEN 'High Risk' WHEN Overrun_Pct > 10 THEN 'Medium Risk'
ELSE 'Low Risk' END AS Risk_Category
```

Project duration derivation

```
DATEDIFF(DAY, Start_Date, Planned_End_Date) AS Planned_Duration
```

6. SQL Views

- **vw_Project_KPIs** — Project-level KPI layer with budget, actual cost, overrun, duration, delay, status, and CASE-based risk category.
- **vw_Cost_Analysis** — Variance-detail layer enriched with project type, region, and project overrun percentage.
- **vw_Equipment_Risk** — Maintenance layer enriched with project type and contractor.

7. Power BI Development

The PBIX contains one report page and uses the three SQL analytical views. The report includes four KPI cards, three slicers, a regional map, a detailed cost-variance table, a donut chart, a monthly line chart, a project-type bar chart, and a contractor/equipment column chart.

The report metadata exposes model measures named Total Planned, Total Actual, Overrun %, and Total Variance SAR. Exact DAX formulas were not documented because the compressed DataModel content was not recoverable during inspection.

8. Dashboard KPIs

KPI	Dashboard value
Total Planned Cost	371.23 bn SAR
Total Actual Cost	482.97 bn SAR
Overrun	30.10%
Total Variance	111.73 bn SAR

9. Key Business Insights

- Aggregate cost pressure is substantial: actual cost is 482.97 bn SAR against 371.23 bn SAR planned, with a displayed 111.73 bn SAR variance and 30.10% overrun.
- The monthly cost trend is volatile, with February at about 84.2M and August at about 97.6M.
- The project-type visual ranges from about 6.0bn to 7.6bn; Residential Building is the highest displayed category and School the lowest.
- The contractor/equipment visual ranges from about 2.6K to 3.2K, with Nesma & Partners highest among the displayed contractors.
- Material Price Escalation appears prominently in the visible variance detail, including two positive variances of roughly 10.25M and 9.72M.

10. Business Recommendations

- Prioritize projects with high overrun percentages for budget review and corrective action.
- Investigate material-price escalation as a recurring cost-variance driver and strengthen procurement/contingency monitoring.
- Use project-type and regional views to focus management attention on the largest displayed cost concentrations.

- Use equipment downtime results from the SQL analysis to prioritize maintenance planning for assets with the highest accumulated downtime.
- Review the dashboard's field/title mismatches before using it as a formal executive reporting artifact.

11. Results

The project demonstrates an end-to-end analytical workflow: source-table validation → SQL business analysis → reusable views → Power BI visualization → executive-level KPI interpretation.

12. Dashboard QA Observations

The PBIX metadata reveals several title-to-field mismatches that should be corrected in a production version: the visual titled “Actual Cost by Project Type” queries Cost_Overrun_SAR; “Cost By Project Type” queries Downtime_Hours; and “Maintenance Cost by Severity” queries a count of Equipment by Contractor. The package preserves the supplied dashboard rather than silently changing its logic.

13. Conclusion

The dashboard communicates a clear budget-pressure story while the SQL layer provides reusable project, variance, and equipment-risk views. The strongest portfolio value is the demonstrated connection between relational SQL analysis and executive Power BI reporting, with the QA observations making the project documentation transparent and technically defensible.

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This report accompanies the full SQL scripts, Power BI (.pbix) file, source datasets, and README documentation available in the project repository.

14. Dashboard Screenshot

Full-resolution capture of the Power BI report referenced throughout this document.



Source: supplied Power BI dashboard screenshot.